

Francisco Rau

AI Engineer

+31 6 38859519 | franciscorau08@gmail.com | franciscorau.com | linkedin.com/in/francisco-rau | Tilburg, NL

Summary

Builder-analyst (MSc Data Science & Society, BSc Economics — Tilburg) who has shipped AI products end to end for the past year: a production SaaS with an AI receptionist (80k lines TypeScript, 294 tests, solo), a player-valuation platform with calibrated ML (23,353 players live), and operational automation at Netvice. Trainee title, shipping habit. Everything runs live at franciscorau.com.

Skills

Data & BI: Python (pandas, scikit-learn), SQL (PostgreSQL), Excel, dashboarding (Streamlit, custom React), ETL & data quality

Modelling: forecasting, regression, gradient boosting, probability calibration, out-of-sample validation

Ways of working: stakeholder requirements, KPI definition, self-directed delivery, AI/agent dev tooling

Experience

Data & Process Analyst — Netvice

May 2025 – Jul 2026 · Tilburg, Netherlands

- Grew from data intern to Data & Process Analyst, working with increasingly senior stakeholders — ultimately on strategic projects directly with the company's founder.
- Established operational BI as a new, self-owned workstream — a first in the company's ten-year history — with full ownership from roadmap to daily use on the call-center floor.
- Fully self-directed: each of the systems below began as my own proposal — need identified, pitched, built, and shipped at pace.
- Forecasting model (the centerpiece): turns planned mailing volume into the exact number of agents needed on any future day — a cohort-based simulation of call attempts, pickup decay, and callback flows. Because it runs off the mailing calendar, agent demand is known 500+ days ahead, far beyond the notice period scheduling requires. Includes capacity-choke detection and a batch-splitting optimizer.
- Performance analytics: a percentile-based hourly pacing metric that makes agents comparable across campaigns with very different call lengths and success rates, combined with outcomes into a single performance index powering real-time TV rankings on the floor.
- Real-time operations layer: live dashboards wired to telephony webhooks (agent line state, idle detection, coverage views), plus a Daily Agent Planner, agent day reports, and a finance board (pilot).
- Introduced agentic AI development (Replit, Claude Code) as a delivery method — an early bet that let one analyst design, build, and ship production tooling end to end.
- Built everything configurable rather than hard-coded, so the tooling keeps working as campaigns and staff change.

Student Researcher — Philips

Sep 2022 – Dec 2022 · Eindhoven, Netherlands

- Researched OneBlade market penetration among students; built linear models estimating adoption of a blade-subscription service across student segments.

Data Entry — Von der Heide

Jan 2019 – Aug 2019 · Mexico City, Mexico

- Cleaned and migrated the company database to a new CRM system, safeguarding data integrity throughout the transfer.

Selected Projects

Market-Gap Analysis: Online-Booking Adoption in Mexico City

2020 · Google Places API, headless browser, Streamlit

- End-to-end data pipeline measuring online-booking adoption across 10,605 businesses: 722 structured API queries, deduplication, two independent booking detectors, and public-registry enrichment. Measured 6.9% adoption; delivered a 9,878-row qualified dataset (89% with direct contact details) as a workbook and dashboard.

- Weather Prediction-Market Research**

2026 · Python, probability calibration, market microstructure
- Tested whether prediction markets price official weather observations efficiently: implied-probability extraction from order books, no-arbitrage coherence checks, repricing-latency measurement. Documented an honest null result.
- Citania — Booking & Payments Platform**

2025–2026 · TypeScript, PostgreSQL, React Native
- Solo-built multi-tenant booking and payments SaaS: 100+ integration tests, database-level slot exclusivity, two-way Google Calendar sync. Live in production.
- Football Player Value Modelling**

2024 · Python, statistical modelling
- Predicted market values on a multi-year, 100+ competition dataset compiled myself. Best model: $R^2 = 0.921$, RMSE \$2.23M.

Education

- MSc Data Science & Society (Business Track) — Tilburg University**

2023 – 2024
- BSc Economics — Tilburg University**

2020 – 2023

Languages

English (native), Spanish (native), Dutch (A2, actively learning)